FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Solids of Revolution

Worked solutions 11 questions · 123 marks

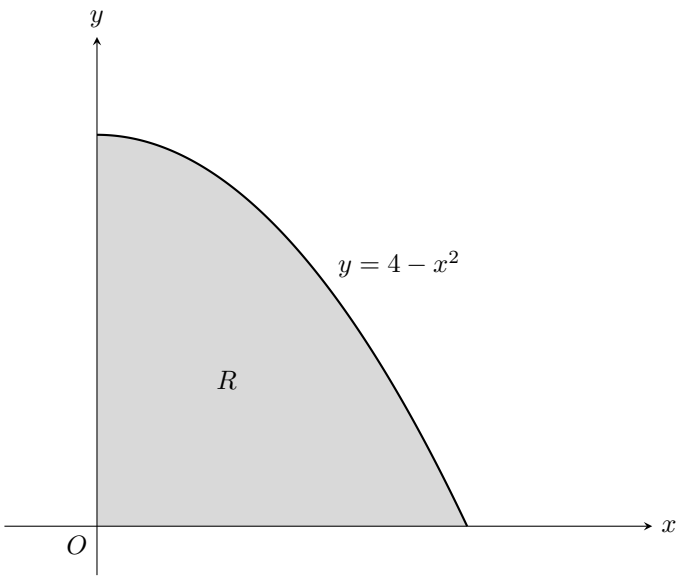
EXAM BOARDS Edexcel FM2 9FM0/4C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Major Y421

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*8 marks*	2
Question 2	*9 marks*	4
Question 3	*12 marks*	6
Question 4	*11 marks*	8
Question 5	*11 marks*	10
Question 6	*14 marks*	12
Question 7	*10 marks*	14
Question 8	*12 marks*	16
Question 9	*12 marks*	18
Question 10	*11 marks*	20
Question 11	*13 marks*	22



1. The shaded region R, shown in the diagram above, is bounded by the curve with equation $y = 4 - x^2$, the positive x-axis and the positive y-axis. A uniform solid S is formed by rotating R through 360° about the x-axis.
Use algebraic integration to find the exact distance of the centre of mass of S from its plane face. **[8]**

Solution

The curve meets the positive x-axis where $4 - x^2 = 0$, that is at $x = 2$, so the region runs from $x = 0$ to $x = 2$. The solid is symmetric about the x-axis, so its centre of mass lies on the x-axis, at $(\bar{x}, 0)$. The solid is uniform, so its density cancels. For a solid formed by rotating a region about the x-axis, the volume and centre of mass satisfy

$$V = \pi \int_a^b y^2 \, \mathrm{d}x, \qquad V\bar{x} = \pi \int_a^b xy^2 \, \mathrm{d}x$$

First expand y^2:

$$y^2 = \left(4 - x^2\right)^2 = 16 - 8x^2 + x^4$$

The volume is

$$\begin{aligned} V &= \pi \int_0^2 \left(16 - 8x^2 + x^4\right) \mathrm{d}x \\ &= \pi \left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_0^2 \\ &= \pi \left(32 - \frac{64}{3} + \frac{32}{5}\right) \\ &= \frac{256\pi}{15} \end{aligned}$$

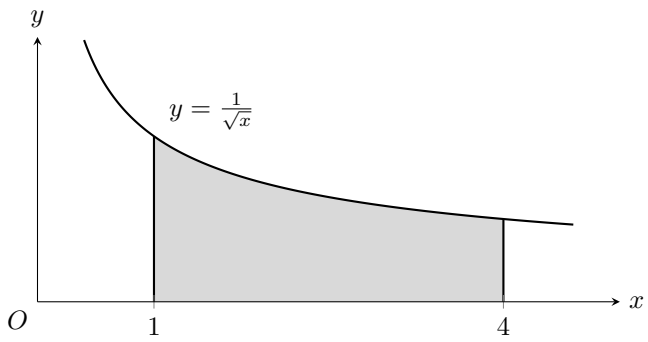
The moment is

$$\begin{aligned}\pi\int_0^2 xy^2\,\mathrm{d}x &= \pi\int_0^2 \left(16x-8x^3+x^5\right)\mathrm{d}x\\ &= \pi\left[8x^2-2x^4+\frac{x^6}{6}\right]_0^2\\ &= \pi\left(32-32+\frac{64}{6}\right)\\ &= \frac{32\pi}{3}\end{aligned}$$

Therefore

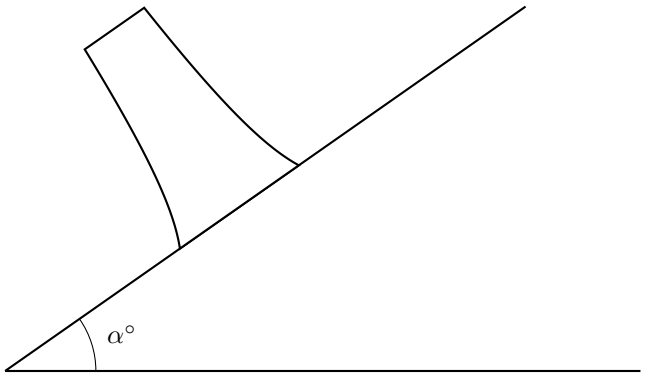
$$\bar{x} = \frac{32\pi}{3}\div\frac{256\pi}{15} = \frac{32\times 15}{3\times 256} = \frac{5}{8}$$

The plane face of S lies in the plane $x=0$, so the centre of mass of S is at a distance $\frac{5}{8}$ from its plane face.



2. The shaded region shown in the diagram above is bounded by the curve with equation $y = \frac{1}{\sqrt{x}}$, the line with equation $x = 1$, the line with equation $x = 4$ and the x-axis.
The region is rotated through 360° about the x-axis to form a solid.
This solid is used to model a uniform solid stand of height 3 m whose plane faces have radii 1 m and 0.5 m.
The larger plane face is the base of the stand.

(a) Show that the centre of mass of the stand is $\left(\frac{3}{2\ln 2} - 1\right)$ m from its base. **[6]**



The stand is placed with its base in contact with a rough plane that is inclined at an angle $\alpha°$ to the horizontal, as shown in the diagram above. The plane is sufficiently rough to prevent the stand from sliding.
Given that the stand is on the point of toppling,

(b) find the value of α, giving your answer to 1 decimal place. **[3]**

Solution

(a) By symmetry, the centre of mass lies on the x-axis. The solid is uniform, so its density cancels. For a solid formed by rotating a region about the x-axis, the volume and centre of mass satisfy

$$V = \pi \int_a^b y^2 \, \mathrm{d}x, \qquad V\bar{x} = \pi \int_a^b xy^2 \, \mathrm{d}x$$

Here $y^2 = \frac{1}{x}$. Since $x > 0$ for $1 \leq x \leq 4$, the logarithm needs no modulus signs. The volume is

$$\begin{aligned} V &= \pi \int_1^4 \frac{1}{x}\,\mathrm{d}x \\ &= \pi[\ln x]_1^4 \\ &= \pi \ln 4 \\ &= 2\pi \ln 2 \end{aligned}$$

The moment is

$$\begin{aligned} \pi \int_1^4 x \times \frac{1}{x}\,\mathrm{d}x &= \pi \int_1^4 1\,\mathrm{d}x \\ &= \pi[x]_1^4 \\ &= 3\pi \end{aligned}$$

Hence

$$\bar{x} = \frac{3\pi}{2\pi \ln 2} = \frac{3}{2\ln 2}$$

The base is the larger plane face, of radius 1 m, which lies in the plane $x = 1$. So the distance of the centre of mass from the base is

$$\left(\frac{3}{2\ln 2} - 1\right) \text{m}$$

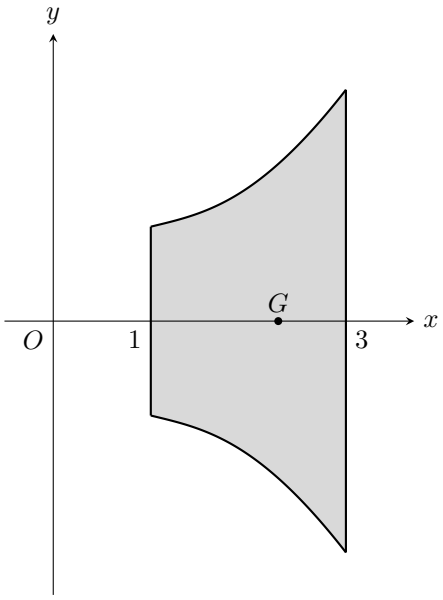
as required.

(b) On the point of toppling, the centre of mass G of the stand lies vertically above the lowest point of the rim of the base. The axis of the stand is perpendicular to the plane, so it makes an angle α° with the vertical. In the right-angled triangle formed by the centre of the base, G and the lowest point of the rim, the side along the axis is $\left(\frac{3}{2\ln 2} - 1\right)$ m and the side along the base is the radius 1 m, so

$$\begin{aligned} \tan \alpha^\circ &= \frac{1}{\frac{3}{2\ln 2} - 1} \\ &= \frac{2\ln 2}{3 - 2\ln 2} \\ &\approx 0.8591 \end{aligned}$$

Therefore

$$\alpha = \tan^{-1}\left(\frac{2\ln 2}{3 - 2\ln 2}\right) \approx 40.7 \qquad (1 \text{ d.p.})$$



3. The region bounded by the curve $y = \sqrt{x^3 - 3x^2 + 4x}$, the x-axis and the lines $x = 1$ and $x = 3$ is rotated through 2π radians about the x-axis to form a uniform solid of revolution R.
The centre of mass of R is the point G, as shown in the diagram above.

(a) Explain why the y-coordinate of G is 0. **[1]**

(b) Find the x-coordinate of G, giving your answer as an exact fraction. **[8]**

P is a point on the circumference of the smaller plane face of R. The solid R is freely suspended from P and hangs in equilibrium.

(c) Find the angle between the axis of symmetry of R and the vertical, giving your answer in degrees to 1 decimal place. **[3]**

Solution

(a) The solid R is formed by rotating about the x-axis and is uniform, so it is symmetric about the x-axis. Hence G lies on the x-axis, and its y-coordinate is 0.

(b) The solid is uniform, so its density cancels. For a solid formed by rotating a region about the x-axis, the volume and centre of mass satisfy

$$V = \pi \int_a^b y^2 \, \mathrm{d}x, \qquad V\bar{x} = \pi \int_a^b xy^2 \, \mathrm{d}x$$

Here $y^2 = x^3 - 3x^2 + 4x$. The volume is

$$\begin{aligned} V &= \pi \int_1^3 \left(x^3 - 3x^2 + 4x\right) \mathrm{d}x \\ &= \pi \left[\frac{x^4}{4} - x^3 + 2x^2\right]_1^3 \\ &= \pi \left[\left(\frac{81}{4} - 27 + 18\right) - \left(\frac{1}{4} - 1 + 2\right)\right] \\ &= \pi \left(\frac{45}{4} - \frac{5}{4}\right) \\ &= 10\pi \end{aligned}$$

The moment is

$$\begin{aligned} \pi \int_1^3 xy^2 \, \mathrm{d}x &= \pi \int_1^3 \left(x^4 - 3x^3 + 4x^2\right) \mathrm{d}x \\ &= \pi \left[\frac{x^5}{5} - \frac{3x^4}{4} + \frac{4x^3}{3}\right]_1^3 \\ &= \pi \left[\left(\frac{243}{5} - \frac{243}{4} + 36\right) - \left(\frac{1}{5} - \frac{3}{4} + \frac{4}{3}\right)\right] \\ &= \pi \left(\frac{477}{20} - \frac{47}{60}\right) \\ &= \frac{346\pi}{15} \end{aligned}$$

Hence

$$\bar{x} = \frac{\frac{346\pi}{15}}{10\pi} = \frac{346}{150} = \frac{173}{75}$$

(c) The smaller plane face is at $x = 1$, where $y^2 = 1 - 3 + 4 = 2$, so its radius is $\sqrt{2}$. Let C be the centre of this face, $C(1, 0)$.

When R hangs in equilibrium from P, G lies vertically below P. The triangle PCG has a right angle at C, with

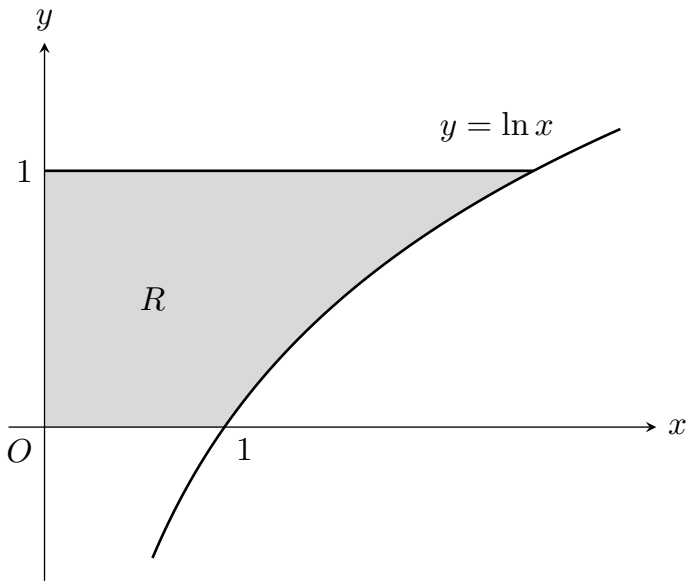
$$CP = \sqrt{2}, \qquad CG = \frac{173}{75} - 1 = \frac{98}{75}$$

The angle θ between the axis of symmetry and the vertical is angle PGC, so

$$\tan\theta = \frac{CP}{CG} = \frac{\sqrt{2}}{\frac{98}{75}} = \frac{75\sqrt{2}}{98}$$

Therefore

$$\theta = \tan^{-1}\left(\frac{75\sqrt{2}}{98}\right) \approx 47.3^\circ \qquad (1 \text{ d.p.})$$



4. The diagram above shows the region R bounded by the curve with equation $y = \ln x$, the line $y = 1$, the x-axis and the y-axis.
A uniform solid S is formed by rotating R through 2π radians about the y-axis.

(a) Show that the volume of S is $\frac{\pi}{2}\left(e^2 - 1\right)$. **[4]**

(b) Find, in terms of e, the distance of the centre of mass of S from O. **[7]**

Solution

(a) The solid is formed by rotating R about the y-axis, so its volume is

$$V = \pi \int_a^b x^2 \, \mathrm{d}y$$

On the curve, $y = \ln x$, so $x = e^y$ and $x^2 = e^{2y}$. The region runs from $y = 0$ to $y = 1$, so

$$\begin{aligned} V &= \pi \int_0^1 e^{2y} \, \mathrm{d}y \\ &= \pi \left[\tfrac{1}{2} e^{2y}\right]_0^1 \\ &= \pi \left(\tfrac{1}{2} e^2 - \tfrac{1}{2}\right) \\ &= \frac{\pi}{2}\left(e^2 - 1\right) \end{aligned}$$

as required.

(b) The solid is symmetric about the y-axis, so its centre of mass lies on the y-axis at $(0, \bar{y})$, and its distance from O is $\bar{y}$. The solid is uniform, so its density cancels. For a solid formed by rotating a region about the y-axis, the centre of mass satisfies

$$V\bar{y} = \pi \int_a^b yx^2 \, \mathrm{d}y$$

With $x^2 = e^{2y}$, and y running from 0 to 1, this becomes

$$V\bar{y} = \pi \int_0^1 y\, e^{2y}\, \mathrm{d}y$$

Integration by parts is

$$\int u \frac{\mathrm{d}v}{\mathrm{d}y}\, \mathrm{d}y = uv - \int v \frac{\mathrm{d}u}{\mathrm{d}y}\, \mathrm{d}y$$

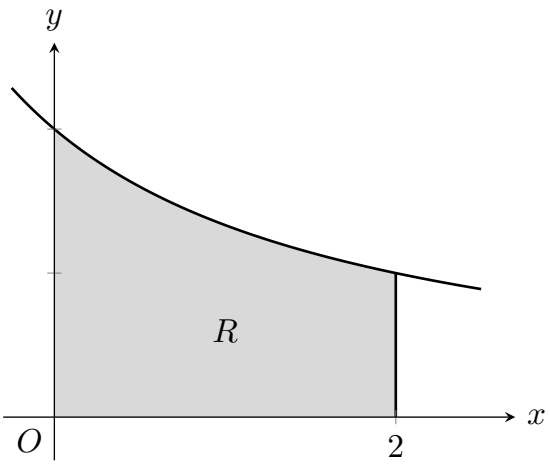
Take $u = y$ and $\frac{\mathrm{d}v}{\mathrm{d}y} = e^{2y}$, so $\frac{\mathrm{d}u}{\mathrm{d}y} = 1$ and $v = \frac{1}{2}e^{2y}$. Then

$$\begin{aligned}
\int_0^1 y\, e^{2y}\, \mathrm{d}y &= \left[\tfrac{1}{2} y\, e^{2y}\right]_0^1 - \int_0^1 \tfrac{1}{2} e^{2y}\, \mathrm{d}y \\
&= \left[\tfrac{1}{2} y\, e^{2y} - \tfrac{1}{4} e^{2y}\right]_0^1 \\
&= \left(\tfrac{1}{2} e^2 - \tfrac{1}{4} e^2\right) - \left(0 - \tfrac{1}{4}\right) \\
&= \frac{e^2 + 1}{4}
\end{aligned}$$

So $V\bar{y} = \frac{\pi}{4}\left(e^2 + 1\right)$. Using part (a):

$$\begin{aligned}
\bar{y} &= \frac{\frac{\pi}{4}\left(e^2 + 1\right)}{\frac{\pi}{2}(e^2 - 1)} \\
&= \frac{e^2 + 1}{2(e^2 - 1)} \approx 0.657
\end{aligned}$$

The centre of mass of S is a distance $\frac{e^2+1}{2(e^2-1)}$ from O.



5. The diagram above shows the region R bounded by the curve with equation $y = \frac{4}{2+x}$, the x-axis, the y-axis and the line $x = 2$.
The region R is rotated through 360° about the x-axis to form a uniform solid.

(a) Show that the centre of mass of the solid is at a distance $4(1 - \ln 2)$ from its smaller plane face. **[8]**

(b) The solid is freely suspended from a point on the circumference of its smaller plane face and hangs in equilibrium.
Find the angle between the axis of symmetry of the solid and the vertical, giving your answer to the nearest degree. **[3]**

Solution

(a) The solid is symmetric about the x-axis, so its centre of mass lies on the x-axis. The solid is uniform, so its density cancels. For a solid formed by rotating a region about the x-axis, the volume and centre of mass satisfy

$$V = \pi \int_a^b y^2 \, \mathrm{d}x, \qquad V\bar{x} = \pi \int_a^b xy^2 \, \mathrm{d}x$$

Here $y^2 = \frac{16}{(2+x)^2}$, so the volume is

$$\begin{aligned} V &= \pi \int_0^2 \frac{16}{(2+x)^2} \, \mathrm{d}x \\ &= 16\pi \left[-\frac{1}{2+x} \right]_0^2 \\ &= 16\pi \left(-\frac{1}{4} + \frac{1}{2} \right) \\ &= 4\pi \end{aligned}$$

To find the moment, write the integrand in partial fractions:

$$\frac{x}{(2+x)^2} = \frac{A}{2+x} + \frac{B}{(2+x)^2} \quad \Longrightarrow \quad x = A(2+x) + B$$

Comparing coefficients of x gives $A = 1$, and setting $x = -2$ gives $B = -2$, so

$$\frac{x}{(2+x)^2} = \frac{1}{2+x} - \frac{2}{(2+x)^2}$$

Since $2 + x > 0$ for $0 \le x \le 2$, the logarithm needs no modulus signs:

$$\begin{aligned}
\pi \int_0^2 \frac{16x}{(2+x)^2}\,\mathrm{d}x &= 16\pi \int_0^2 \left(\frac{1}{2+x} - \frac{2}{(2+x)^2}\right)\mathrm{d}x \\
&= 16\pi \left[\ln(2+x) + \frac{2}{2+x}\right]_0^2 \\
&= 16\pi \left[\left(\ln 4 + \frac{1}{2}\right) - (\ln 2 + 1)\right] \\
&= 16\pi \left(\ln 2 - \frac{1}{2}\right) \\
&= \pi(16\ln 2 - 8)
\end{aligned}$$

Hence

$$\bar{x} = \frac{\pi(16\ln 2 - 8)}{4\pi} = 4\ln 2 - 2$$

At $x = 0$ the radius is $y = 2$, and at $x = 2$ it is $y = 1$, so the smaller plane face lies in the plane $x = 2$. The distance of the centre of mass from the smaller face is

$$2 - (4\ln 2 - 2) = 4 - 4\ln 2 = 4(1 - \ln 2)$$

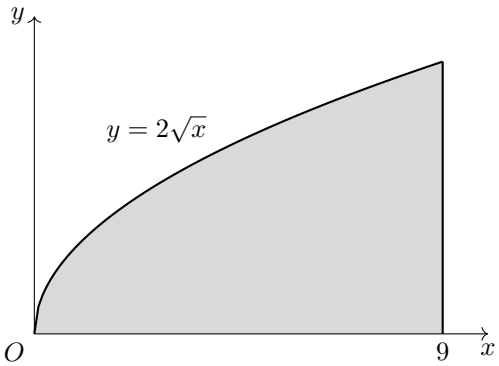
as required.

(b) Let P be the point of suspension, C the centre of the smaller face and G the centre of mass. In equilibrium, G lies vertically below P. Triangle PCG has a right angle at C, with $CP = 1$ (the radius of the smaller face) and $CG = 4(1 - \ln 2)$ from part (a). The angle θ between the axis CG and the vertical PG satisfies

$$\tan\theta = \frac{CP}{CG} = \frac{1}{4(1-\ln 2)}$$

so

$$\theta = \tan^{-1}\left(\frac{1}{4(1-\ln 2)}\right) \approx 39^\circ \qquad \text{(nearest degree)}$$



6. The region bounded by the curve $y = 2\sqrt{x}$, the x-axis and the line $x = 9$, shown in the diagram above, is rotated through 360° about the x-axis to form a solid with a circular plane face.
All units are in centimetres.

(a) State an assumption that you should make about the solid in order to find the position of its centre of mass. **[1]**

(b) Using integration, show that the centre of mass of the solid is 3 cm from its plane face. **[6]**

(c) The solid is placed with its plane face on a rough board. One end of the board is lifted so that the angle between the board and the horizontal is gradually increased.

(i) Assuming that the solid does not slide, find the angle between the board and the horizontal when the solid is on the point of toppling, giving your answer to the nearest degree. **[3]**

(ii) The coefficient of friction between the solid and the board is 1.5. Determine whether the solid first slides or first topples as the angle is increased, justifying your answer. **[4]**

Solution

(a) The solid is uniform: its density is constant throughout.

(b) By symmetry, the centre of mass lies on the x-axis. The solid is uniform, so its density cancels. For a solid formed by rotating a region about the x-axis, the volume and centre of mass satisfy

$$V = \pi \int_a^b y^2 \, \mathrm{d}x, \qquad V\bar{x} = \pi \int_a^b xy^2 \, \mathrm{d}x$$

The volume is

$$\begin{aligned} V &= \pi \int_0^9 y^2 \, \mathrm{d}x \\ &= \pi \int_0^9 4x \, \mathrm{d}x \\ &= \pi \left[2x^2\right]_0^9 \\ &= 162\pi \, \mathrm{cm}^3 \end{aligned}$$

The moment is

$$\pi \int_0^9 xy^2 \,\mathrm{d}x = \pi \int_0^9 4x^2 \,\mathrm{d}x$$
$$= \pi \left[\tfrac{4}{3}x^3\right]_0^9$$
$$= 972\pi$$

so

$$\bar{x} = \frac{972\pi}{162\pi} = 6$$

The plane face is at $x = 9$, so the centre of mass is $9 - 6 = 3\,\text{cm}$ from the plane face, as required.

(c) (i) The plane face is a circle of radius $2\sqrt{9} = 6\,\text{cm}$. The centre of mass is on the axis of symmetry, 3 cm from the face, so the line from it perpendicular to the board meets the board at the centre of the face. On the point of toppling, the centre of mass is vertically above the lowest point of the rim of the plane face. The angle between this vertical and the axis of symmetry equals the angle θ between the board and the horizontal, so

$$\tan\theta = \frac{6}{3} = 2 \quad \Longrightarrow \quad \theta = \tan^{-1} 2 \approx 63^\circ \qquad \text{(nearest degree)}$$

(ii) Let the mass of the solid be m, the normal reaction R and the friction force F, and let the board make angle θ with the horizontal.

Resolving perpendicular to the board:

$$R = mg\cos\theta$$

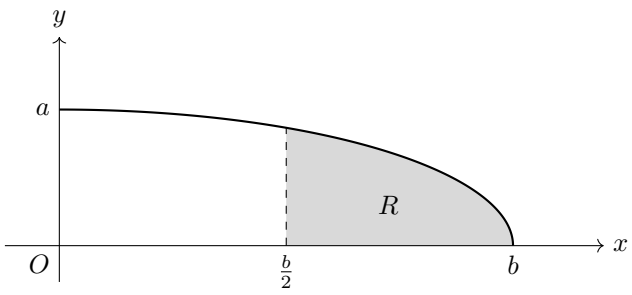
Resolving along the board:

$$F = mg\sin\theta$$

The solid is on the point of sliding when $F = \mu R$ with $\mu = 1.5$, which gives

$$mg\sin\theta = 1.5\, mg\cos\theta$$
$$\tan\theta = 1.5$$
$$\theta \approx 56.3^\circ$$

The solid would slide at $\theta \approx 56.3^\circ$ but would topple only at $\theta \approx 63.4^\circ$. Since $56.3^\circ < 63.4^\circ$, the solid slides first.



7. The diagram above shows the region R bounded by the curve with equation

$$y = \frac{a}{b}\sqrt{b^2 - x^2}$$

the x-axis and the line $x = \frac{b}{2}$, where a and b are positive constants.
The region R is rotated through 360° about the x-axis to form a uniform solid S.

(a) Use integration to show that the volume of S is $\frac{5}{24}\pi a^2 b$. **[4]**

(b) Use integration to show that the centre of mass of S is at a distance $\frac{7b}{40}$ from its plane face. **[6]**

Solution

(a) The volume of revolution about the x-axis is

$$V = \pi \int_{b/2}^{b} y^2 \, \mathrm{d}x$$

where $y^2 = \frac{a^2}{b^2}\left(b^2 - x^2\right)$, so

$$\begin{aligned} V &= \frac{\pi a^2}{b^2} \int_{b/2}^{b} \left(b^2 - x^2\right) \mathrm{d}x \\ &= \frac{\pi a^2}{b^2} \left[b^2 x - \frac{x^3}{3} \right]_{b/2}^{b} \\ &= \frac{\pi a^2}{b^2} \left[\left(b^3 - \frac{b^3}{3} \right) - \left(\frac{b^3}{2} - \frac{b^3}{24} \right) \right] \\ &= \frac{\pi a^2}{b^2} \left(\frac{16b^3}{24} - \frac{11b^3}{24} \right) \\ &= \frac{5}{24}\pi a^2 b \end{aligned}$$

as required.

(b) By symmetry, the centre of mass lies on the x-axis. The solid is uniform, so its density cancels. For a solid formed by rotating a region about the x-axis, the centre of mass satisfies

$$V\bar{x} = \pi \int_{b/2}^{b} x y^2 \, \mathrm{d}x$$

The volume is $\frac{5}{24}\pi a^2 b$ from part (a). The moment is

$$\begin{aligned}
\pi \int_{b/2}^{b} xy^2 \,\mathrm{d}x &= \frac{\pi a^2}{b^2} \int_{b/2}^{b} \left(b^2 x - x^3\right) \mathrm{d}x \\
&= \frac{\pi a^2}{b^2} \left[\frac{b^2 x^2}{2} - \frac{x^4}{4}\right]_{b/2}^{b} \\
&= \frac{\pi a^2}{b^2} \left[\left(\frac{b^4}{2} - \frac{b^4}{4}\right) - \left(\frac{b^4}{8} - \frac{b^4}{64}\right)\right] \\
&= \frac{\pi a^2}{b^2} \left(\frac{16b^4}{64} - \frac{7b^4}{64}\right) \\
&= \frac{9}{64}\pi a^2 b^2
\end{aligned}$$

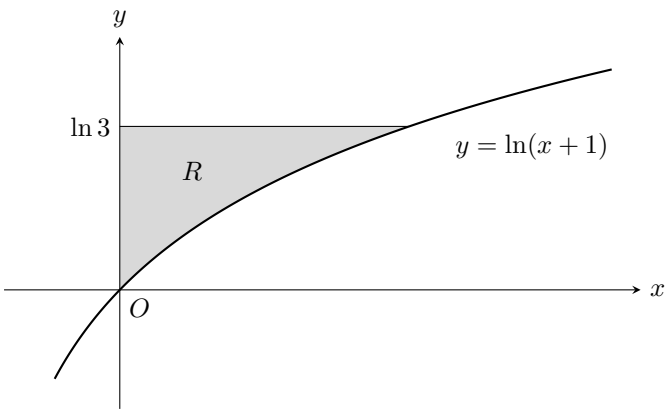
Therefore

$$\bar{x} = \frac{9}{64}\pi a^2 b^2 \times \frac{24}{5\pi a^2 b} = \frac{27b}{40}$$

The plane face of S lies in the plane $x = \frac{b}{2}$. The distance of the centre of mass from this face is

$$\frac{27b}{40} - \frac{b}{2} = \frac{27b}{40} - \frac{20b}{40} = \frac{7b}{40}$$

as required.



8. The diagram above shows part of the curve with equation $y = \ln(x+1)$. The region R, shown shaded in the diagram above, is bounded by the curve, the y-axis and the line $y = \ln 3$.
A uniform solid is formed by rotating the region R through 2π radians about the y-axis.

(a) Show that the volume of the solid is $\pi \ln 3$. **[4]**

(b) Find the exact y-coordinate of the centre of mass of the solid. **[8]**

Solution

(a) Rearrange the equation of the curve to give x in terms of y:

$$y = \ln(x+1) \quad \Longrightarrow \quad x + 1 = e^y \quad \Longrightarrow \quad x = e^y - 1$$

The volume of revolution about the y-axis is

$$V = \pi \int_a^b x^2 \, \mathrm{d}y$$

Here $x^2 = (e^y - 1)^2 = e^{2y} - 2e^y + 1$, and y runs from 0 to $\ln 3$, so

$$\begin{aligned} V &= \pi \int_0^{\ln 3} \left(e^{2y} - 2e^y + 1\right) \mathrm{d}y \\ &= \pi \left[\tfrac{1}{2}e^{2y} - 2e^y + y\right]_0^{\ln 3} \\ &= \pi \left[\left(\tfrac{9}{2} - 6 + \ln 3\right) - \left(\tfrac{1}{2} - 2 + 0\right)\right] \\ &= \pi \left[\left(\ln 3 - \tfrac{3}{2}\right) + \tfrac{3}{2}\right] \\ &= \pi \ln 3 \end{aligned}$$

as required.

(b) The solid is formed by rotation about the y-axis, so its centre of mass lies on the y-axis. The solid is uniform, so its density cancels. For a solid formed by rotating a region about the y-axis, the centre of mass satisfies

$$V\bar{y} = \pi \int_a^b yx^2 \, \mathrm{d}y$$

The volume is $\pi \ln 3$ from part (a). The moment is

$$\pi \int_0^{\ln 3} y\big(e^{2y} - 2e^y + 1\big)\mathrm{d}y = \pi\left(\int_0^{\ln 3} ye^{2y}\,\mathrm{d}y - 2\int_0^{\ln 3} ye^y\,\mathrm{d}y + \int_0^{\ln 3} y\,\mathrm{d}y\right)$$

The first two integrals use integration by parts, which is

$$\int u\frac{\mathrm{d}v}{\mathrm{d}y}\,\mathrm{d}y = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}y}\,\mathrm{d}y$$

For $\int ye^{2y}\,\mathrm{d}y$, take $u = y$, $\frac{\mathrm{d}v}{\mathrm{d}y} = e^{2y}$, so $\frac{\mathrm{d}u}{\mathrm{d}y} = 1$, $v = \frac{1}{2}e^{2y}$:

$$\begin{aligned}\int_0^{\ln 3} ye^{2y}\,\mathrm{d}y &= \left[\tfrac{1}{2}ye^{2y}\right]_0^{\ln 3} - \int_0^{\ln 3} \tfrac{1}{2}e^{2y}\,\mathrm{d}y \\ &= \left[\tfrac{1}{2}ye^{2y} - \tfrac{1}{4}e^{2y}\right]_0^{\ln 3} \\ &= \left(\tfrac{9}{2}\ln 3 - \tfrac{9}{4}\right) - \left(0 - \tfrac{1}{4}\right) \\ &= \tfrac{9}{2}\ln 3 - 2\end{aligned}$$

For $\int ye^{y}\,\mathrm{d}y$, take $u = y$, $\frac{\mathrm{d}v}{\mathrm{d}y} = e^{y}$, so $\frac{\mathrm{d}u}{\mathrm{d}y} = 1$, $v = e^{y}$:

$$\begin{aligned}\int_0^{\ln 3} ye^{y}\,\mathrm{d}y &= \left[ye^{y}\right]_0^{\ln 3} - \int_0^{\ln 3} e^{y}\,\mathrm{d}y \\ &= \left[ye^{y} - e^{y}\right]_0^{\ln 3} \\ &= (3\ln 3 - 3) - (0 - 1) \\ &= 3\ln 3 - 2\end{aligned}$$

The last integral is

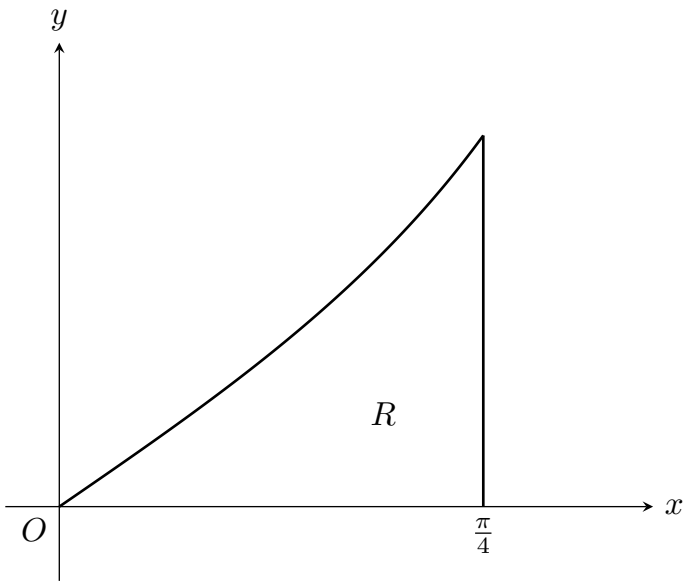
$$\int_0^{\ln 3} y\,\mathrm{d}y = \left[\tfrac{1}{2}y^2\right]_0^{\ln 3} = \tfrac{1}{2}(\ln 3)^2$$

Combining these:

$$\begin{aligned}\pi\int_0^{\ln 3} yx^2\,\mathrm{d}y &= \pi\left[\left(\tfrac{9}{2}\ln 3 - 2\right) - 2(3\ln 3 - 2) + \tfrac{1}{2}(\ln 3)^2\right] \\ &= \pi\big(2 - \tfrac{3}{2}\ln 3 + \tfrac{1}{2}(\ln 3)^2\big)\end{aligned}$$

Dividing by the volume:

$$\begin{aligned}\bar{y} &= \frac{\pi\big(2 - \frac{3}{2}\ln 3 + \frac{1}{2}(\ln 3)^2\big)}{\pi \ln 3} \\ &= \frac{4 - 3\ln 3 + (\ln 3)^2}{2\ln 3} \approx 0.870\end{aligned}$$



9. The diagram above shows the finite region R which is bounded by part of the curve with equation $y = \tan x$, the x-axis and the line with equation $x = \frac{\pi}{4}$.
A uniform solid S is formed by rotating R through 2π radians about the x-axis.
Using algebraic integration,

(a) show that the volume of S is $\frac{\pi}{4}(4 - \pi)$, **[4]**

(b) find the exact x-coordinate of the centre of mass of S. **[8]**

Solution

(a) The volume of a solid formed by rotating a region about the x-axis is

$$V = \pi \int_a^b y^2 \, \mathrm{d}x$$

With $y = \tan x$, and using $\tan^2 x = \sec^2 x - 1$:

$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{4}} \tan^2 x \, \mathrm{d}x \\ &= \pi \int_0^{\frac{\pi}{4}} \left(\sec^2 x - 1\right) \mathrm{d}x \\ &= \pi [\tan x - x]_0^{\frac{\pi}{4}} \\ &= \pi \left(1 - \frac{\pi}{4}\right) - \pi(0 - 0) \\ &= \frac{\pi}{4}(4 - \pi) \end{aligned}$$

as required.

(b) The solid is symmetric about the x-axis, so its centre of mass lies on the x-axis. The solid is uniform, so its density cancels. For a solid formed by rotating a region about the x-axis, the centre of mass satisfies

$$V\bar{x} = \pi \int_a^b xy^2 \, \mathrm{d}x$$

To find the moment, use $\tan^2 x = \sec^2 x - 1$ again:

$$\pi \int_0^{\frac{\pi}{4}} x \tan^2 x \, \mathrm{d}x = \pi \int_0^{\frac{\pi}{4}} x \sec^2 x \, \mathrm{d}x - \pi \int_0^{\frac{\pi}{4}} x \, \mathrm{d}x$$

Integration by parts is

$$\int u \frac{\mathrm{d}v}{\mathrm{d}x} \, \mathrm{d}x = uv - \int v \frac{\mathrm{d}u}{\mathrm{d}x} \, \mathrm{d}x$$

Take $u = x$, $\frac{\mathrm{d}v}{\mathrm{d}x} = \sec^2 x$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$, $v = \tan x$. The standard result is

$$\int \tan x \, \mathrm{d}x = \ln |\sec x| + c$$

Since $\sec x > 0$ for $0 \leq x \leq \frac{\pi}{4}$, the modulus is not needed:

$$\begin{aligned} \int x \sec^2 x \, \mathrm{d}x &= x \tan x - \int \tan x \, \mathrm{d}x \\ &= x \tan x - \ln(\sec x) + c \end{aligned}$$

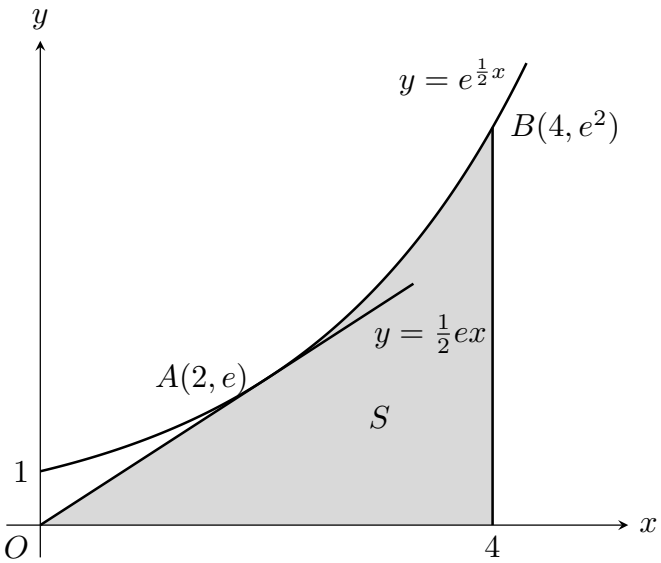
Hence

$$\begin{aligned} \pi \int_0^{\frac{\pi}{4}} x \tan^2 x \, \mathrm{d}x &= \pi \left[x \tan x - \ln(\sec x) - \frac{x^2}{2} \right]_0^{\frac{\pi}{4}} \\ &= \pi \left(\frac{\pi}{4} \times 1 - \ln \sqrt{2} - \frac{\pi^2}{32} \right) - \pi(0 - \ln 1 - 0) \\ &= \pi \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 - \frac{\pi^2}{32} \right) \\ &= \frac{\pi}{32} \left(8\pi - 16 \ln 2 - \pi^2 \right) \end{aligned}$$

Using part (a), $V = \frac{\pi}{4}(4 - \pi) = \frac{8\pi}{32}(4 - \pi)$, so

$$\begin{aligned} \bar{x} &= \frac{\frac{\pi}{32}\left(8\pi - 16 \ln 2 - \pi^2 \right)}{\frac{8\pi}{32}(4 - \pi)} \\ &= \frac{8\pi - 16 \ln 2 - \pi^2}{8(4 - \pi)} \approx 0.608 \end{aligned}$$

10. In this question you may use the facts that for a uniform solid right circular cone of height h and base radius r the volume is $\frac{1}{3}\pi r^2 h$ and the centre of mass is $\frac{1}{4}h$ above the base on the line from the centre of the base to the vertex.



The diagram above shows the shaded region S bounded by the x-axis, the line $y = \frac{1}{2}ex$ for $0 \leq x \leq 2$, the curve $y = e^{\frac{1}{2}x}$ for $2 \leq x \leq 4$, and the line $x = 4$.
The line $y = \frac{1}{2}ex$ is the tangent to the curve $y = e^{\frac{1}{2}x}$ at the point A with coordinates $(2, e)$. The point B on the curve has coordinates $(4, e^2)$.
The region S is rotated through 2π radians about the x-axis to form a uniform solid of revolution T.

(a) Show that the x-coordinate of the centre of mass of T is

$$\frac{9e^2}{3e^2 - 1}$$

[8]

Solid T is freely suspended from B and hangs in equilibrium.

(b) Find the angle between the axis of symmetry of T and the vertical. **[3]**

Solution

(a) The solid T is made of two pieces: a cone, formed by rotating the line $y = \frac{1}{2}ex$ for $0 \leq x \leq 2$, and a second solid, formed by rotating the curve $y = e^{\frac{1}{2}x}$ for $2 \leq x \leq 4$.

The cone. It has vertex O, height 2 and base radius e, with its base in the plane $x = 2$. Its volume is

$$V_1 = \frac{1}{3}\pi e^2 \times 2 = \frac{2\pi e^2}{3}$$

Its centre of mass is $\frac{1}{4} \times 2 = \frac{1}{2}$ from the base towards the vertex O, so $x_1 = 2 - \frac{1}{2} = \frac{3}{2}$.

The second solid. Here $y^2 = e^x$. Using the volume of revolution and its moment:

$$V_2 = \pi \int_2^4 y^2 \, \mathrm{d}x, \qquad V_2 x_2 = \pi \int_2^4 xy^2 \, \mathrm{d}x$$

The volume is

$$V_2 = \pi \int_2^4 e^x \,\mathrm{d}x = \pi[e^x]_2^4 = \pi\big(e^4 - e^2\big)$$

For this moment, integration by parts is

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

with $u = x$, $\frac{\mathrm{d}v}{\mathrm{d}x} = e^x$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = 1$, $v = e^x$. Then

$$\begin{aligned}\int_2^4 xe^x \,\mathrm{d}x &= [xe^x]_2^4 - \int_2^4 e^x \,\mathrm{d}x \\ &= [(x-1)e^x]_2^4 \\ &= 3e^4 - e^2\end{aligned}$$

so

$$V_2 x_2 = \pi\big(3e^4 - e^2\big)$$

The whole solid. For a body made of pieces, the centre of mass satisfies

$$M\bar{x} = \sum m_i x_i, \qquad \text{where } M = \sum m_i$$

The solid is uniform, so masses are proportional to volumes, and $V\bar{x} = \sum V_i x_i$, where $V = \sum V_i$.

The total volume is

$$V = \frac{2\pi e^2}{3} + \pi\big(e^4 - e^2\big) = \pi e^4 - \frac{\pi e^2}{3} = \frac{\pi e^2\big(3e^2 - 1\big)}{3}$$

and the total moment is

$$V_1 x_1 + V_2 x_2 = \frac{2\pi e^2}{3} \times \frac{3}{2} + \pi\big(3e^4 - e^2\big) = \pi e^2 + 3\pi e^4 - \pi e^2 = 3\pi e^4$$

Therefore

$$\bar{x} = \frac{3\pi e^4}{\frac{\pi e^2\big(3e^2 - 1\big)}{3}} = \frac{9e^2}{3e^2 - 1}$$

as required.

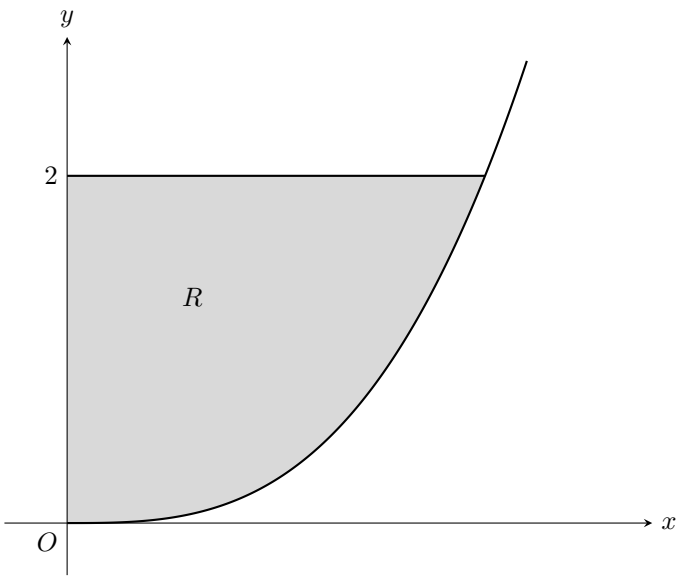
(b) When T hangs freely from B, its centre of mass G lies vertically below B. Let C be the centre of the plane face, at $(4, 0)$. Then B lies on the rim of this face, so BC is perpendicular to the axis with $BC = e^2$, and G lies on the axis with

$$GC = 4 - \frac{9e^2}{3e^2 - 1} = \frac{12e^2 - 4 - 9e^2}{3e^2 - 1} = \frac{3e^2 - 4}{3e^2 - 1}$$

The angle θ between the axis and the vertical is the angle BGC in the right-angled triangle BGC, so

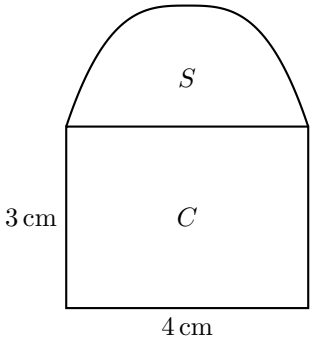
$$\tan\theta = \frac{BC}{GC} = \frac{e^2\big(3e^2 - 1\big)}{3e^2 - 4} \approx 8.609$$

$$\theta \approx 83.4^\circ$$



11. The region R, shown shaded in the diagram above, is bounded by part of the curve with equation $y = \frac{1}{4}x^3$, where $x \geq 0$, the line with equation $y = 2$ and the y-axis. The unit of length on both axes is one centimetre.
A uniform solid, S, is formed by rotating R through 360° about the y-axis.

(a) Find the volume of S, giving your answer as an exact multiple of π, and show that the centre of mass of S is $\frac{3}{4}$ cm from its plane face. **[6]**



A uniform solid cylinder, C, has base radius 2 cm and height 3 cm. The cylinder C is attached to S so that the plane face of S coincides with a plane face of C, forming the paperweight P shown in the diagram above. The density of the material used to make S is k times the density of the material used to make C. The plane face of P rests on a desk lid that is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{4}{5}$. The lid is sufficiently rough to prevent P from slipping.

(b) Find the greatest value of k for which P can rest in equilibrium on the lid without toppling. **[7]**

Solution

(a) The solid is uniform, so its density cancels. For a solid formed by rotating a region about the y-axis, the volume and centre of mass satisfy

$$V = \pi \int_a^b x^2 \, \mathrm{d}y, \qquad V\bar{y} = \pi \int_a^b yx^2 \, \mathrm{d}y$$

From $y = \frac{1}{4}x^3$, $x^3 = 4y$, so

$$x^2 = (4y)^{\frac{2}{3}} = 2^{\frac{4}{3}} y^{\frac{2}{3}}$$

The region runs from $y = 0$ to $y = 2$, so

$$\begin{aligned} V &= \pi \int_0^2 2^{\frac{4}{3}} y^{\frac{2}{3}} \, \mathrm{d}y \\ &= 2^{\frac{4}{3}} \pi \left[\tfrac{3}{5} y^{\frac{5}{3}} \right]_0^2 \\ &= \tfrac{3}{5} \pi \times 2^{\frac{4}{3}} \times 2^{\frac{5}{3}} \\ &= \tfrac{3}{5} \pi \times 2^3 \\ &= \frac{24\pi}{5} \, \mathrm{cm}^3 \end{aligned}$$

The moment is

$$\begin{aligned} \pi \int_0^2 y x^2 \, \mathrm{d}y &= \pi \int_0^2 2^{\frac{4}{3}} y^{\frac{5}{3}} \, \mathrm{d}y \\ &= 2^{\frac{4}{3}} \pi \left[\tfrac{3}{8} y^{\frac{8}{3}} \right]_0^2 \\ &= \tfrac{3}{8} \pi \times 2^{\frac{4}{3}} \times 2^{\frac{8}{3}} \\ &= \tfrac{3}{8} \pi \times 2^4 \\ &= 6\pi \end{aligned}$$

so

$$\bar{y} = \frac{6\pi}{\frac{24\pi}{5}} = \frac{30}{24} = \frac{5}{4}$$

The centre of mass is on the y-axis, $\frac{5}{4}$ cm above O. The plane face of S lies in the plane $y = 2$, so the distance of the centre of mass from the plane face is

$$2 - \frac{5}{4} = \frac{3}{4} \, \mathrm{cm}$$

as required.

(b) Let the density of the material of C be ρ, so that the density of S is $k\rho$. Measure distances along the axis of symmetry of P from the plane face of P (the base of C).

The cylinder C has volume $\pi \times 2^2 \times 3 = 12\pi \, \mathrm{cm}^3$, so its mass is $12\pi\rho$, and its centre of mass is $\frac{3}{2}$ cm from the base.

The solid S has mass $\frac{24\pi k\rho}{5}$, and by part (a) its centre of mass is $3 + \frac{3}{4} = \frac{15}{4}$ cm from the base.

By symmetry, the centre of mass G of P lies on the axis; let it be $\bar{x}$ cm from the base. For a body made of pieces, the centre of mass satisfies

$$M\bar{x} = \sum m_i x_i, \qquad \text{where } M = \sum m_i$$

so

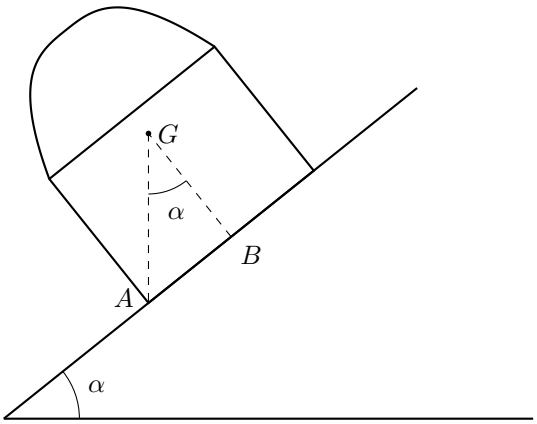
$$\left(12\pi\rho + \frac{24\pi k\rho}{5} \right) \bar{x} = 12\pi\rho \times \frac{3}{2} + \frac{24\pi k\rho}{5} \times \frac{15}{4}$$

Dividing by $\pi\rho$:

$$\left(12 + \frac{24k}{5}\right)\bar{x} = 18 + 18k$$

$$\bar{x} = \frac{90 + 90k}{60 + 24k} = \frac{15(1+k)}{10 + 4k}$$

The diagram below shows P on the point of toppling, where B is the centre of the base and A is the lowest point of the rim of the base.



Since BG is perpendicular to the lid and GA is vertical, angle $BGA = \alpha$. P does not topple provided the vertical through G meets the lid within the base, that is, provided angle BGA is at least α, so we need

$$\bar{x}\tan\alpha \leq 2$$

$$\frac{4}{5}\bar{x} \leq 2$$

$$\bar{x} \leq \frac{5}{2}$$

Substituting for $\bar{x}$, with $10 + 4k > 0$:

$$\frac{15(1+k)}{10+4k} \leq \frac{5}{2}$$

$$30 + 30k \leq 50 + 20k$$

$$10k \leq 20$$

$$k \leq 2$$

So the greatest value of k is 2.